

TRANSPORT AND TILING BOUNDS FOR THE AFFINE PLANK PROBLEM ON THE TRIANGLE

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ABSTRACT. Bang’s affine (relative-width) plank conjecture asserts that if a convex body K is covered by planks $P_1, \dots, P_n$ then $\sum_i \text{rw}_K(P_i) \geq 1$, where $\text{rw}_K(P) = w(P)/w_K(u_P)$ is the width of P relative to the width of K in P ’s direction. It is open for non-symmetric bodies, and open already for three planks in the plane. We study the triangle as a concrete test body. We give (i) a short single-measure proof of $\sum \text{rw} \geq 1/d$ for $d \geq 2$, whose per-plank estimate is sharp at the simplex, so that no single-measure argument can improve on $1/d$; (ii) sharp $\sum \text{rw} \geq 1$ for two directions and for facet-parallel families (all dimensions); (iii) a Hunter-free proof of the “three facets + one arbitrary plank” case; and, as our main results, (iv) what is, to our knowledge, the first *tight non-facet* case with full rigidity: three planks parallel to the medians of a triangle satisfy $\sum \text{rw} \geq 1$ with equality if and only if each is the central third of its median; and (v) a *weighted-perimeter theorem* extending the bound to a two-parameter family of generic concurrent direction triples, via an explicit witness measure with edge masses $1 - 2p_j$. We isolate the transport method behind (i)–(v) as a *uniform-marginal witness measure* and give a dimension-agnostic necessary condition for its existence (concurrence of the mid-level lines); for cyclic triples the condition is also sufficient, giving a complete characterization. Finally we record a structural obstruction: the chord-lemma step in the sign argument of Bang, which yields the chord-normalized bound $\sum w(P_i)/\ell_K(u_i) \geq 1$, is sharp, so that argument cannot be pushed past the longest chord: no normalizer strictly between chord and width is reachable by it. We claim no proof of the conjecture.

1. INTRODUCTION

A *plank* of width w is the region between two parallel hyperplanes at distance w . For a convex body $K \subset \mathbb{R}^d$ and a plank P with unit normal u , the *relative width* of P is $\text{rw}_K(P) = w(P)/w_K(u)$, where $w_K(u) = \max_{x \in K} \langle u, x \rangle - \min_{x \in K} \langle u, x \rangle$. Bang’s affine plank conjecture [3, 1] states:

Conjecture 1.1 (Affine plank problem; Bang 1951). If K is covered by planks $P_1, \dots, P_n$, then $\sum_{i=1}^n \text{rw}_K(P_i) \geq 1$.

Ball [2] proved Conjecture 1.1 for centrally symmetric K . For general bodies it is open, and, as emphasised by Bakaev–Yehudayoff [4], it is open *already in the plane* — only the two-plank case is settled there. The best general lower bound is $\sum \text{rw} \geq 2/(1 + \sqrt{d})$ [4]; on the equilateral triangle their method gives $\sum \text{rw} \geq 4\sqrt{3} - 6 \approx 0.928$.

Scope. This paper studies the triangle as a concrete body and develops the associated *transport* (witness-measure) and *tiling* methods. It is *not* an attack on Conjecture 1.1 through the Ambrus reduction: that reduction sends a covering by N planks to a simplex of dimension $2N - 1$, so the triangle (the 2-simplex) is never a target (Remark 1.2). Our results are new sharp cases for a fixed body, a rigidity theorem, a dimension-agnostic structural characterization, and an obstruction; we prove nothing about the general conjecture.

Remark 1.2. Ambrus’s reduction [1] sends a covering by N planks to a covering of a simplex of dimension $2N - 1$ (two points per plank). The target simplices thus have odd dimension growing with N ; the 2-simplex (triangle) is never among them. So proving the conjecture for the triangle does *not* imply the planar case, and our results are about the triangle as a concrete body, not a reduction of Conjecture 1.1.

The triangle model. By affine invariance take $\Delta = \{x \in \mathbb{R}^3 : x_i \geq 0, x_1 + x_2 + x_3 = 1\}$. A normalized plank is $P = \{x \in \Delta : f(x) \in I\}$ with $f : \Delta \rightarrow [0, 1]$ affine onto $[0, 1]$ and $I \subset [0, 1]$ an interval; then $\text{rw}_\Delta(P) = |I|$. Covering means $\Delta \subset \bigcup_i P_i$. The restriction $I \subset [0, 1]$ loses no generality: replacing I by $I \cap [0, 1]$ leaves $P \cap \Delta$ unchanged and does not increase $|I|$.

Results. Theorem 2.1 (§2), Theorems 3.1 and 3.2 (§3), Theorem 4.1 (§4), and the centerpieces: Theorem 5.1 with rigidity (§5) and the weighted-perimeter Theorem 6.3 (§6), which extends the sharp bound from the medians to a two-parameter family of generic concurrent triples. Proposition 6.1 (§6) is the dimension-agnostic necessary condition, and Corollary 6.4 the resulting characterization for cyclic triples; Proposition 7.1 (§7) is the normalizer obstruction.

2. THE SINGLE-MEASURE BOUND $\sum \text{rw} \geq 1/d$

Theorem 2.1. *Let $d \geq 2$. If $K \subset \mathbb{R}^d$ is covered by planks $P_1, \dots, P_n$, then $\sum_i \text{rw}_K(P_i) \geq 1/d$. The per-plank estimate $\mu(P) \leq d \text{rw}_K(P)$ used in the proof is sharp at the simplex; hence no single-measure argument can improve the constant $1/d$. (For $d = 1$ the conjecture itself is trivial: intervals covering an interval satisfy $\sum \text{rw} \geq 1$.)*

Proof. Let μ be the uniform probability measure on K . Fix u , normalize $\langle u, \cdot \rangle \in [0, w]$, $w = w_K(u)$. The marginal density is $f(t) = V(t)/\text{vol}(K)$ with $V(t) = \text{vol}_{d-1}(K \cap \{\langle u, \cdot \rangle = t\})$; by Brunn–Minkowski $g := V^{1/(d-1)}$ is concave on $[0, w]$. If g peaks at t^* with value g^* , then $g \geq g^*h$ for the tent h ($h(0) = h(w) = 0$, $h(t^*) = 1$), so $f = g^{d-1} \geq (\max f) h^{d-1}$ and $1 = \int f \geq (\max f) \int_0^w h^{d-1} = (\max f) w/d$, giving $\max f \leq d/w$. Hence $\mu(P) = \int_I f \leq d \text{rw}_K(P)$ for each plank, and $1 = \mu(K) \leq \sum_i \mu(P_i) \leq d \sum_i \text{rw}_K(P_i)$. For sharpness of the per-plank estimate: equality in $\max f \leq d/w$ forces $V \propto h^{d-1}$ (conical sections), realized by the simplex $K = \Delta^d$ with u a vertex–facet direction, where a thin plank at the facet has $\mu(P) = (d - o(1)) \text{rw}_K(P)$. Thus for the simplex no single probability measure can certify a constant larger than $1/d$ by the union bound; this does *not* assert any covering of the simplex with $\sum \text{rw}$ near $1/d$ (indeed Theorem 3.2 gives $\sum \text{rw} \geq 1$ for facet-parallel coverings). $\square$

Remark 2.2. The bound cannot exceed $1/d$ by a single measure: uniform marginals in all directions would force $\mathbb{E}[x_k] = 1/2$ against $\sum x_k = 1$. This is Gardner’s obstruction [5]: for the triangle there is no relative-width measure for the three side-directions.

3. SHARP CASES: TWO DIRECTIONS AND FACET-PARALLEL FAMILIES

Theorem 3.1. *If every plank covering the triangle is parallel to one of two fixed non-parallel directions, then $\sum \text{rw} \geq 1$, and this is sharp. (If the two directions are parallel the statement is trivial: all planks share one normalized coordinate f , the intervals $f(P_i)$ cover $[0, 1]$, and $\sum \text{rw} = \sum |I_i| \geq 1$.)*

Proof. Let $f_u, f_v : \Delta \rightarrow [0, 1]$ be the two affine coordinates; since the directions are non-parallel, $\Phi = (f_u, f_v)$ is an affine bijection onto a triangle $T \subset [0, 1]^2$ with full projections. By [5, Theorem 1], which constructs a relative-width measure for any *two* directions on any planar convex body, there is a probability measure ν on Δ with $f_{u\#}\nu = f_{v\#}\nu = \text{Leb}$. Then $\nu(P) = |I| = \text{rw}(P)$ for each plank, and $1 = \nu(\Delta) \leq \sum \nu(P_i) = \sum \text{rw}(P_i)$. $\square$

Theorem 3.2. *Planks parallel to the facets of a simplex Δ^d , in any number, that cover Δ^d satisfy $\sum \text{rw} \geq 1$, and this is sharp.*

Proof. Write $\Delta^d = \{\lambda \in [0, 1]^{d+1} : \sum_k \lambda_k = 1\}$ in barycentric coordinates; a plank parallel to facet k is a coordinate slab $\{\lambda_k \in I\}$. Grouping the planks by facet, let $U_k = \bigcup_{c(i)=k} I_i \subset [0, 1]$ be the forbidden set on facet k , so $\sum_k |U_k| \leq \sum_i |I_i| =: R$, and set $C_k = [0, 1] \setminus U_k$. A point $\lambda \in \Delta^d$ is uncovered iff $\lambda_k \in C_k$ for every k ; since $C_k \subset [0, 1]$, an uncovered point exists iff $1 \in C_1 + \dots + C_{d+1}$ (Minkowski sum).

Assume $R < 1$; we produce an uncovered point. Each C_k is nonempty compact ($|C_k| = 1 - |U_k| \geq 1 - R > 0$). Put $D = C_1 + \dots + C_d$. By the one-dimensional Brunn–Minkowski inequality $|A+B| \geq |A|+|B|$ (for nonempty compact $A, B \subset \mathbb{R}$, iterated), $|D| \geq \sum_{k=1}^d |C_k| = d - \sum_{k=1}^d |U_k|$; as $D \subseteq [0, d]$, $|[0, 1] \setminus D| \leq |[0, d] \setminus D| \leq \sum_{k=1}^d |U_k|$, whence $|D \cap [0, 1]| \geq 1 - \sum_{k=1}^d |U_k|$. Also $(1 - C_{d+1}) \cap [0, 1]$ has measure $|C_{d+1}| = 1 - |U_{d+1}|$. Both sets lie in $[0, 1]$ and their measures sum to $2 - \sum_k |U_k| \geq 2 - R > 1$, so they intersect: there is $t \in D$ with $1 - t \in C_{d+1}$. Then $t + (1 - t) = 1 \in C_1 + \dots + C_{d+1}$, i.e. an uncovered λ exists. Contrapositively, covering forces $R \geq 1$. Sharpness: the facet tiling $I_k = [0, \frac{1}{d+1}]$ gives $\sum \text{rw} = 1$. $\square$

Remark 3.3. The edge-by-edge argument does *not* prove this: on an edge only two barycentric coordinates vary, so no single edge can force $\sum_k |U_k| \geq 1$ across all $d+1$ facets. The Minkowski-sum reduction above is the correct route; note that no witness *measure* can prove it either, since facet directions violate the concurrence condition of Proposition 6.1.

4. THREE FACETS PLUS ONE ARBITRARY PLANK

Theorem 4.1. *Let $P_i = \{\lambda_i \in I_i\}$ ($i = 1, 2, 3$) be facet-parallel with $a_i = |I_i|$, and let $Q = \{f \in J\}$, $t = |J|$, with $f : \Delta \rightarrow [0, 1]$ affine normalized. If $\Delta \subset P_1 \cup P_2 \cup P_3 \cup Q$ then $a_1 + a_2 + a_3 + t \geq 1$.*

Proof. Choose an active edge of f ; relabelling, $f(v_j) = 0$, $f(v_k) = 1$, and let v_i be the opposite vertex, $\theta = f(v_i)$. For $s \in [0, 1]$ consider the fibre $F_s = \{\lambda_i = s\}$, parametrized by $u = \lambda_k \in [0, 1 - s]$, so $\lambda_j = 1 - s - u$ and $f|_{F_s}(u) = \theta s + u$. On F_s : $P_i \cap F_s = F_s$ if $s \in I_i$ (else $\emptyset$); $P_j \cap F_s$, $P_k \cap F_s$, $Q \cap F_s$ have lengths $\leq a_j, a_k, t$. Put $B = a_j + a_k + t$. If $s \notin I_i$ and F_s is covered, its length $1 - s \leq B$. If $B \geq 1$ we are done; else $[0, 1 - B] \subset I_i$, so $a_i \geq 1 - B$. In both cases $a_1 + a_2 + a_3 + t = a_i + B \geq 1$. $\square$

5. THE MEDIAN THEOREM AND ITS RIGIDITY

Let m_1, m_2, m_3 be the medians of Δ as normalized affine forms: with A, B, C the vertices,

$$m_1 = (\tfrac{1}{2}, 0, 1), \quad m_2 = (0, 1, \tfrac{1}{2}), \quad m_3 = (1, \tfrac{1}{2}, 0),$$

(vertex values). One has $m_1 + m_2 + m_3 \equiv \frac{3}{2}$. A *median plank* is $P_i = \{m_i \in I_i\}$, $I_i = [l_i, h_i]$, $r_i = |I_i| > 0$.

Theorem 5.1. *If three median planks cover Δ , then $r_1 + r_2 + r_3 \geq 1$, with equality if and only if $I_1 = I_2 = I_3 = [\frac{1}{3}, \frac{2}{3}]$.*

To our knowledge this is the first tight *non-facet* case of the three-plank problem with full rigidity: the sharp cases recorded in the survey [6] (and in [5, 4]) concern facet/coordinate directions, two directions, or symmetric bodies, and we are not aware of a comparable tilted tight instance in the literature.

Proof of the bound. Let μ be the perimeter measure: uniform in the parameter on each of the three edges, mass $\frac{1}{3}$ each. A direct computation shows each pushforward $m_{i\#}\mu = \text{Leb}[0, 1]$. Indeed for m_1 : on AB , m_1 ranges over $[0, \frac{1}{2}]$ with Jacobian 2 (density $\frac{1}{3} \cdot 2$); on CA , over $[\frac{1}{2}, 1]$

likewise; on BC , over $[0, 1]$ with Jacobian 1 (density $\frac{1}{3}$); the total density is 1 on $[0, 1]$. Hence $\mu(P_i) = \text{Leb}(I_i) = r_i$, and $1 = \mu(\Delta) \leq \sum \mu(P_i) \leq \sum r_i$. $\square$

Proof of rigidity. Equality in $1 \leq \sum \mu(P_i)$ forces the P_i to partition μ a.e.; since μ lives on the edges, on each edge the three traces tile $[0, 1]$. We claim there are exactly three such edge-tilings, and only one covers the interior.

Three edge-tilings. On each edge each median is affine, so the three traces are explicit intervals depending piecewise-linearly on (l_i, h_i) through $a_i = |I_i \cap [0, \frac{1}{2}]|$ and $b_i = |I_i \cap [\frac{1}{2}, 1]|$. Classifying each I_i by crossing type (contained in $[0, \frac{1}{2}]$, crossing $\frac{1}{2}$, or contained in $[\frac{1}{2}, 1]$) and each edge by the left-to-right order of its traces, the abutting (tiling) conditions become a finite family of linear systems in (l_i, h_i) .

Lemma 5.2. *The edge-tiling conditions have exactly three solutions with $0 \leq l_i < h_i \leq 1$: $[\frac{1}{3}, \frac{2}{3}]^3$, $[0, \frac{1}{3}]^3$, and $[\frac{2}{3}, 1]^3$.*

A complete, self-contained proof is given in Appendix A: the symmetries of the median configuration reduce the a priori 27 crossing patterns to seven classes, five of which are impossible and two of which lead to 3×3 linear systems with the stated unique solutions. The lemma has also been verified by three independent exact computer enumerations, archived as ancillary files (`median_rigidity_enumeration.py`, `median_edgetilings_independent.py`, `median_edgetiling_types_crosscheck.py`).

Interior witness. The centroid G has $m(G) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Since $\frac{1}{2} \notin [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, the tilings $[0, \frac{1}{3}]^3$ and $[\frac{2}{3}, 1]^3$ leave G uncovered; they tile the boundary but are not coverings of Δ . The tiling $[\frac{1}{3}, \frac{2}{3}]^3$ covers Δ and has $\sum r_i = 1$. Hence the unique covering with $\sum r_i = 1$ is $[\frac{1}{3}, \frac{2}{3}]^3$. $\square$

6. A DIMENSION-AGNOSTIC NECESSARY CONDITION FOR TRANSPORT

The proofs of Theorems 3.1 and 5.1 share one device: a probability measure whose pushforward under each direction is uniform. Which direction-tuples admit one?

Proposition 6.1. *Let $u_1, \dots, u_{d+1}$ be normalized affine forms on Δ^d with vertex-value matrix V invertible. If a probability measure μ on Δ^d has $u_{i\#}\mu = \text{Leb}[0, 1]$ for all i , then its barycenter $p \in \Delta^d$ satisfies $Vp = \frac{1}{2}\mathbf{1}$; equivalently the mid-level hyperplanes $\{u_i = \frac{1}{2}\}$ concur at p , which requires $\mathbf{1}^\top V^{-1}\mathbf{1} = 2$ and $V^{-1}\mathbf{1} \geq 0$.*

Proof. $\frac{1}{2} = \mathbb{E}_\mu[u_i] = \sum_j V_{ij} \mathbb{E}_\mu[x_j] = (Vp)_i$, and $p \in \Delta^d$ gives $\mathbf{1}^\top p = 1$, $p \geq 0$; substitute $p = \frac{1}{2}V^{-1}\mathbf{1}$. $\square$

The three medians satisfy this with p the centroid; the three facets do not ($\mathbf{1}^\top V^{-1}\mathbf{1} = 3 \neq 2$), recovering Gardner's obstruction. The concurrence family is two-dimensional (parametrized by p): a normalized direction has vertex values $\{0, \tau, 1\}$, and $Vp = \frac{1}{2}\mathbf{1}$ ties each τ_i to p . The medians are the case $p = \text{centroid}$, forcing $\tau_i = \frac{1}{2}$. Among these, the explicit perimeter measure singles out the medians:

Proposition 6.2. *For a direction u with vertex values $\{0, \tau, 1\}$, the perimeter measure μ (uniform on each edge, mass $\frac{1}{3}$) satisfies $u_{\#}\mu = \text{Leb}[0, 1]$ if and only if $\tau = \frac{1}{2}$.*

Proof. On the three edges u is affine between the pairs $\{0, \tau\}$, $\{\tau, 1\}$, $\{0, 1\}$, so $u_{\#}\mu$ has density $\frac{1}{3\tau} + \frac{1}{3}$ on $[0, \tau]$ and $\frac{1}{3(1-\tau)} + \frac{1}{3}$ on $[\tau, 1]$. Both equal 1 iff $\tau = \frac{1}{2}$. $\square$

Uniform edge-weights thus single out the medians. Allowing *unequal* edge masses, however, the construction covers a full two-parameter family:

Theorem 6.3 (weighted perimeter). *Let $p = (p_A, p_B, p_C)$ be interior with $\max_j p_j < \frac{1}{2}$, and let $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ (vertex values at A, B, C) be the cyclic directions concurrent at p , i.e. $\tau_1 = \frac{1/2 - p_C}{p_A}$, $\tau_2 = \frac{1/2 - p_B}{p_C}$, $\tau_3 = \frac{1/2 - p_A}{p_B}$. Let μ_p be uniform on each edge with masses $w_{AB} = 1 - 2p_C$, $w_{BC} = 1 - 2p_A$, $w_{CA} = 1 - 2p_B$ (summing to 1). Then $u_i \# \mu_p = \text{Leb}[0, 1]$ for $i = 1, 2, 3$; consequently every covering of Δ by planks in these three directions satisfies $\sum \text{rw} \geq 1$.*

Proof. Each u_i is affine of full range $[0, 1]$ on the edge joining its 0- and 1-vertices (u_1 on BC , u_2 on AB , u_3 on CA) and affine of range $[0, \tau_i]$, resp. $[\tau_i, 1]$, on the other two. The pushforward of a uniform edge mass w_e through an affine map of range length L is uniform with density w_e/L . Hence $u_1 \# \mu_p$ has density $w_{BC} + w_{AB}/\tau_1$ on $[0, \tau_1]$ and $w_{BC} + w_{CA}/(1 - \tau_1)$ on $[\tau_1, 1]$, and similarly for u_2, u_3 : six equations. Using $p_A + p_B + p_C = 1$, each composite term collapses by the identity $(1 - 2x)p/(\frac{1}{2} - x) = 2p$: e.g. $w_{AB}/\tau_1 = (1 - 2p_C)p_A/(\frac{1}{2} - p_C) = 2p_A$, so $w_{BC} + w_{AB}/\tau_1 = (1 - 2p_A) + 2p_A = 1$; likewise $1 - \tau_1 = \frac{1/2 - p_B}{p_A}$ gives $w_{CA}/(1 - \tau_1) = 2p_A$. All six densities equal 1, so each marginal is $\text{Leb}[0, 1]$, and the transport argument of Theorem 5.1 applies verbatim. Finally $\tau_i \in (0, 1)$ for all i iff $\max_j p_j < \frac{1}{2}$ iff all three masses are positive, so every valid cyclic triple admits μ_p . $\square$

Taking p the centroid recovers Theorem 5.1's bound ($\tau_i = \frac{1}{2}$, equal masses). For p off the centroid the directions are generic non-median triples; e.g. $p = (\frac{9}{20}, \frac{3}{10}, \frac{1}{4})$ gives $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$ with masses $(\frac{1}{2}, \frac{1}{10}, \frac{2}{5})$. Theorem 6.3 thus proves the bound of Conjecture 1.1 for a two-parameter family of generic direction triples on the triangle, answering the finite-direction existence question of [5] affirmatively for the cyclic concurrent family.

Combining necessity (Proposition 6.1) with the explicit construction (Theorem 6.3) yields the first *characterization* in this setting:

Corollary 6.4. *Call a direction triple cyclic if its vertex values are $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ with $\tau_i \in (0, 1)$. A cyclic triple admits a uniform-marginal witness measure if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ concur. In that case the concurrence point p automatically lies in the open medial triangle ($p_j > 0$ and $p_j < \frac{1}{2}$ for all j), and the weighted-perimeter measure μ_p of Theorem 6.3 is a witness.*

Proof. The vertex-value matrix has $\det V = -(1 + \tau_1\tau_2\tau_3) \neq 0$, so Proposition 6.1 applies and gives necessity. Conversely, if the lines concur at p (a point of the affine plane of Δ , so $\sum_j p_j = 1$), then p is the unique solution of $Vp = \frac{1}{2}\mathbf{1}$, which in closed form reads $2(1 + \tau_1\tau_2\tau_3)p_A = 1 - \tau_3(1 - \tau_2)$ and cyclically; each right-hand side is positive for $\tau_i \in (0, 1)$, so $p_j > 0$, and the identities $\frac{1}{2} - p_C = \tau_1 p_A$, $\frac{1}{2} - p_A = \tau_3 p_B$, $\frac{1}{2} - p_B = \tau_2 p_C$ give $p_j < \frac{1}{2}$. Theorem 6.3 then supplies μ_p . $\square$

Remark 6.5. The *anti-cyclic* pattern (all 0- and 1-roles exchanged) reduces to the cyclic one by a reflection of Δ (a transposition of two vertices), which maps the medial triangle to itself; Theorem 6.3 and Corollary 6.4 hold for it verbatim. Non-cyclic (mixed) role assignments, and the rigidity of the equality case for p off the centroid, remain open (Problem 8.2).

The concurrence condition is dimension-agnostic: the cyclic medians of Δ^d ($m_i(e_i) = \frac{1}{2}$, $m_i(e_{i+1}) = 0$, $m_i(e_{i-1}) = 1$, else $\frac{1}{2}$) have row-sums $\frac{d+1}{2}$, hence concur at the centroid in every dimension. Whether a witness measure then exists for $d \geq 3$ is open (the perimeter measure is special to $d = 2$); this is a higher-dimensional instance of the finite-direction question left open by Gardner [5].

7. A NORMALIZER OBSTRUCTION

For a direction u let $\ell_K(u)$ be the length of the longest chord of K parallel to u ; always $\ell_K(u) \leq w_K(u)$. Bakaev–Yehudayoff’s chord bound $\sum_i w(P_i)/\ell_K(u_i) \geq 1$ [4] follows from Bang’s sign argument via the chord lemma. Consider interpolating the normalizer, $N_c = (1 - c)\ell + c w$, $c \in [0, 1]$: $N_c = \ell$ at $c = 0$ (chord bound) and $N_c = w$ at $c = 1$ (Conjecture 1.1).

Proposition 7.1. *The chord-lemma step is sharp: for u with $\|u\| = a/2$, the body $(K - u) \cap (K + u)$ contains a homothet of K of ratio $(\ell - a)/\ell$ [6, Lem. 2.3], and this ratio is attained (by a segment). Consequently the iterated chord-lemma placement of a Bang-system witness $x_0 + \sum_j \pm u_j$ proves $\sum_i w(P_i)/N(u_i) \geq 1$ for $N = \ell$ and for no direction-normalizer $N > \ell$; in particular it does not establish $\sum_i w(P_i)/N_c(u_i) \geq 1$ for any $c > 0$.*

Proof. The placement argument runs as follows: iterating the chord lemma over the directions u_j , the common intersection $\bigcap_j ((K - u_j) \cap (K + u_j))$ (in the appropriate nested form) is nonempty as soon as $\sum_j w(P_j)/\ell_j < 1$, which places the witness and forces $\sum w(P_j)/\ell_j \geq 1$ for coverings. The movement budget available in direction u is therefore governed by the homothety ratio $(\ell - a)/\ell$. This ratio is tight: for a segment $K = [y, z]$ of length ℓ parallel to u , $(K - u) \cap (K + u)$ is a segment of length exactly $\ell - a$. Hence the total budget in direction u is exactly the longest chord $\ell(u)$ and cannot be stretched to any $N(u) > \ell(u)$; since $N_c(u) = (1 - c)\ell(u) + c w_K(u) > \ell(u)$ whenever $c > 0$ and $w_K(u) > \ell(u)$, the claim follows. $\square$

Remark 7.2. Proposition 7.1 does not show $\sum w(P_i)/N_c(u_i) \geq 1$ is false for $c > 0$; it shows the sign-argument route cannot prove it. Beating $2/(1 + \sqrt{d})$ by this normalizer therefore requires a witness that is not a Bang-system point, i.e. one exploiting the coupling of the covering. This is consistent with the sharpness of [4] for the cube.

8. OPEN PROBLEMS

Problem 8.1. The three-plank case for three genuinely tilted directions on the triangle (equivalently Conjecture 1.1 for the triangle).

Problem 8.2. Corollary 6.4 characterizes the cyclic (and anti-cyclic) role assignments. For the remaining *mixed* role assignments of a concurrent triple, decide whether a uniform-marginal witness measure exists; and determine the equality (rigidity) case of Theorem 6.3 for p off the centroid — does $\sum rw = 1$ force a tiling, and is it ever attained?

Problem 8.3. Whether $\sum w(P_i)/N_c(u_i) \geq 1$ holds for some $c > 0$ (a coupling certificate), which would improve the general bound past $2/(1 + \sqrt{d})$.

APPENDIX A. PROOF OF LEMMA 5.2

Setup. Parametrize each edge by $t \in [0, 1]$, in the orientations $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$. From the vertex values $m_1 = (\frac{1}{2}, 0, 1)$, $m_2 = (0, 1, \frac{1}{2})$, $m_3 = (1, \frac{1}{2}, 0)$, each edge sees one median with values $[0, \frac{1}{2}]$ (the *low* median c , with $m_c(t) = \frac{1}{2}(1 - t)$), one with values $[0, 1]$ (the *full* median f , $m_f(t) = t$), and one with values $[\frac{1}{2}, 1]$ (the *high* median g , $m_g(t) = 1 - \frac{t}{2}$), with roles

$$(c, f, g) = (1, 2, 3) \text{ on } AB, \quad (3, 1, 2) \text{ on } BC, \quad (2, 3, 1) \text{ on } CA.$$

Inverting, the traces $T_i = \{t : m_i(t) \in I_i\}$ on an edge with roles (c, f, g) are

$$\begin{aligned} T_c &= [\max(0, 1 - 2h_c), 1 - 2l_c] \text{ (nondegenerate iff } l_c < \tfrac{1}{2}), & T_f &= [l_f, h_f], \\ T_g &= [2 - 2h_g, \min(1, 2 - 2l_g)] \text{ (nondegenerate iff } h_g > \tfrac{1}{2}). \end{aligned} \tag{1}$$

Classify each $I_i = [l_i, h_i]$ ($l_i < h_i$) by its position relative to $\frac{1}{2}$: type L if $h_i \leq \frac{1}{2}$, type R if $l_i \geq \frac{1}{2}$, type M if $l_i < \frac{1}{2} < h_i$; every nondegenerate interval is of exactly one type. Two elementary observations, used throughout:

(O1) If I_c is of type M then $T_c = [0, 1 - 2l_c]$ (it contains the left endpoint); if I_c is of type R then T_c is degenerate (a point or empty). Dually, if I_g is of type M then $T_g = [2 - 2h_g, 1]$; if I_g is of type L then T_g is degenerate.

(O2) The tiling condition (traces cover the edge, with pairwise disjoint interiors) is insensitive to degenerate traces: a degenerate trace covers at most one point, and since the nondegenerate traces are finitely many closed sets whose union contains $[0, 1]$ minus finitely many points, that union is all of $[0, 1]$. So on each edge the *nondegenerate* traces tile $[0, 1]$; in particular, if finitely many closed intervals of positive length have disjoint interiors and cover $[0, 1]$, they abut consecutively from 0 to 1.

Symmetries. The rotation $A \rightarrow B \rightarrow C \rightarrow A$ maps the median configuration to itself and acts on solutions by $(I_1, I_2, I_3) \mapsto (I_2, I_3, I_1)$, hence on type vectors by a cyclic shift. The reflection swapping B and C maps $(m_1, m_2, m_3) \rightarrow (1 - m_1, 1 - m_3, 1 - m_2)$, hence acts by $(I_1, I_2, I_3) \mapsto (1 - I_1, 1 - I_3, 1 - I_2)$ (where $1 - [l, h] = [1 - h, 1 - l]$), which exchanges types $L \leftrightarrow R$. Under the group generated by these, the 27 type vectors $(X_1, X_2, X_3) \in \{L, M, R\}^3$ fall into seven orbits, with representatives

$$\text{MMM}; \quad \text{LLL}; \quad \text{MML}; \quad \text{LLM}; \quad \text{LLR}; \quad \text{LMR}; \quad \text{LRM}$$

(of sizes 1, 2, 6, 6, 6, 3, 3). It suffices to analyze one representative per orbit.

Case MMM. By (O1), on every edge $T_c = [0, 1 - 2l_c]$ and $T_g = [2 - 2h_g, 1]$, both nondegenerate, and $T_f = [l_f, h_f]$ has positive length. Disjointness of interiors forces $1 - 2l_c \leq 2 - 2h_g$ with T_f inside the gap, and covering forces equality of the abutting endpoints:

$$l_f = 1 - 2l_c, \quad h_f = 2 - 2h_g.$$

Reading this on AB, BC, CA gives $l_2 = 1 - 2l_1, l_1 = 1 - 2l_3, l_3 = 1 - 2l_2$, whose unique solution is $l_1 = l_2 = l_3 = \frac{1}{3}$, and $h_2 = 2 - 2h_3, h_1 = 2 - 2h_2, h_3 = 2 - 2h_1$, whose unique solution is $h_i = \frac{2}{3}$. This is $[\frac{1}{3}, \frac{2}{3}]^3$, which is indeed of type MMM and tiles each edge as $[0, \frac{1}{3}] \cup [\frac{1}{3}, \frac{2}{3}] \cup [\frac{2}{3}, 1]$.

Case LLL. All T_g are degenerate, so each edge is tiled by $T_c = [1 - 2h_c, 1 - 2l_c]$ (the full image, since $I_c \subset [0, \frac{1}{2}]$) and $T_f = I_f \subset [0, \frac{1}{2}]$. Since $h_f \leq \frac{1}{2} < 1$, the tile containing 1 must be T_c , forcing $l_c = 0$. If $h_c = \frac{1}{2}$ then $T_c = [0, 1]$ and T_f would have empty interior, impossible; so T_f contains 0 ($l_f = 0$) and abuts T_c : $h_f = 1 - 2h_c$. On the three edges: $l_i = 0$ for all i and $h_2 = 1 - 2h_1, h_1 = 1 - 2h_3, h_3 = 1 - 2h_2$, forcing $h_i = \frac{1}{3}$. This is $[0, \frac{1}{3}]^3$ (type LLL, tiling each edge as $[0, \frac{1}{3}] \cup [\frac{1}{3}, 1]$). The reflected orbit representative RRR gives $[\frac{2}{3}, 1]^3$.

Case MML (impossible). On AB : T_3 is degenerate (I_3 of type L), so $[0, 1]$ is tiled by $T_1 = [0, 1 - 2l_1]$ (O1) and $T_2 = I_2$. If $1 \in T_1$ then $T_1 = [0, 1]$ and T_2 has empty interior, impossible; so $h_2 = 1$. Then on BC the high trace (1) is $T_2 = [2 - 2h_2, \min(1, 2 - 2l_2)] = [0, 1]$ (as $h_2 = 1, l_2 < \frac{1}{2}$), leaving the full trace $T_1 = I_1$ (positive length) with empty interior — contradiction.

Case LLM (impossible). On BC the roles are $(c, f, g) = (3, 1, 2)$: the high trace is degenerate (I_2 of type L), so $[0, 1]$ is tiled by $T_3 = [0, 1 - 2l_3]$ (O1, I_3 of type M) and $T_1 = I_1 \subset [0, \frac{1}{2}]$. The tile containing 1 must be T_3 ($h_1 \leq \frac{1}{2}$), so $l_3 = 0$ and $T_3 = [0, 1]$, leaving T_1 with empty interior — contradiction.

Case LLR (impossible). On BC both compressed traces are degenerate (I_3 of type R gives a degenerate low trace, I_2 of type L a degenerate high trace), so $T_1 = I_1$ alone tiles $[0, 1]$, forcing $h_1 = 1 > \frac{1}{2}$ — contradicting type L.

Case LMR (impossible). On BC $(c, f, g) = (3, 1, 2)$: low trace degenerate (I_3 type R), so $T_1 = I_1$ and $T_2 = [2 - 2h_2, 1]$ (O1) tile $[0, 1]$; as above this forces $l_1 = 0$ and $h_1 = 2 - 2h_2$. On CA $(c, f, g) = (2, 3, 1)$: high trace degenerate (I_1 type L), so $T_2 = [0, 1 - 2l_2]$ (O1) and $T_3 = I_3$ tile, forcing $h_3 = 1$ and $l_3 = 1 - 2l_2$. Now on AB : $T_3 = [2 - 2h_3, 2 - 2l_3] = [0, 4l_2]$, which has length $2r_3 > 0$, so $l_2 > 0$; but then the interiors of $T_3 = [0, 4l_2]$ and $T_2 = [l_2, h_2]$ overlap on $(l_2, \min(4l_2, h_2)) \neq \emptyset$ — contradiction.

Case LRM (impossible). On CA $(c, f, g) = (2, 3, 1)$: both compressed traces are degenerate (I_2 of type R, I_1 of type L), so $T_3 = I_3$ alone tiles $[0, 1]$, i.e. $I_3 = [0, 1]$. Then on AB the high trace (1) is $T_3 = [2 - 2h_3, \min(1, 2 - 2l_3)] = [0, 1]$, leaving $T_2 = I_2$ (positive length) with empty interior — contradiction.

All orbits are exhausted: the only solutions are $[\frac{1}{3}, \frac{2}{3}]^3$, $[0, \frac{1}{3}]^3$, $[\frac{2}{3}, 1]^3$. □

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